

Inequalities – 5

Matrix and spectral inequalities

10 сентября 2026 года

Theoretical minimum.

Theorem 1 (the scalar product of matrices). $\langle A, B \rangle = \operatorname{tr}(AB^*)$ is an inner product, its norm is the Frobenius norm $\|A\|_F^2 = \sum |a_{ij}|^2 = \sum \sigma_i^2$, and Cauchy–Bunyakovsky gives $|\operatorname{tr}(AB^*)| \leq \|A\|_F \|B\|_F$. Von Neumann’s trace inequality sharpens this to $|\operatorname{tr}(AB)| \leq \sum_i \sigma_i(A) \sigma_i(B)$.

Theorem 2 (Hadamard and Minkowski for determinants). For any matrix $|\det A| \leq \prod_j \|A_{\cdot j}\|_2$, with equality exactly for orthogonal columns. For positive definite A and B the function $A \mapsto \log \det A$ is concave, so $\det(tA + (1-t)B) \geq (\det A)^t (\det B)^{1-t}$, and $\det(A+B)^{1/n} \geq (\det A)^{1/n} + (\det B)^{1/n}$.

Fact 1 (eigenvalues move slowly). For symmetric matrices the eigenvalues in decreasing order satisfy Weyl’s inequalities $|\lambda_k(A+E) - \lambda_k(A)| \leq \|E\|$ and the Hoffman–Wielandt bound $\sum_k |\lambda_k(A) - \lambda_k(B)|^2 \leq \|A - B\|_F^2$; the variational description $\lambda_k = \min_{\dim L=k} \max_{x \in L} \frac{\langle Ax, x \rangle}{\langle x, x \rangle}$ is what makes all of them work.

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1. Prove Theorem 1 in full, including von Neumann’s trace inequality, and prove Hadamard’s inequality together with its equality case.
 2. (MIPT 2014) Let A and B be linear operators on a finite-dimensional Euclidean space such that $\|Ax\| \leq \|Bx\|$ for every vector x . Prove that $|\det A| \leq |\det B|$. Deduce the monotonicity of the determinant on the positive semidefinite cone: if $0 \preceq A \preceq B$, then $\det A \leq \det B$.
 3. Prove the Ky Fan maximum principle: for a symmetric A the sum of the k largest eigenvalues equals $\max \{\operatorname{tr}(P^\top A P)\}$ over all $n \times k$ matrices P .

with orthonormal columns. Deduce that this sum is a convex function of A , and prove Weyl's inequality $\lambda_{i+j-1}(A+B) \leq \lambda_i(A) + \lambda_j(B)$ for symmetric A and B .

4. Prove the Schur product theorem: the entrywise product of two positive semidefinite matrices is positive semidefinite. Deduce that $\operatorname{tr}(AB) \leq \operatorname{tr} A \cdot \operatorname{tr} B$ for positive semidefinite A, B , and that $\det(A+B) \geq \det A + \det B$.
5. (VJIMC 2014) Let $0 < a < b$ and let $f: [a, b] \rightarrow \mathbb{R}$ be continuous with $\int_a^b f = 0$. Show that $\int_a^b \int_a^b f(x)f(y) \ln(x+y) \, dx \, dy \leq 0$, and explain the statement as the negative semidefiniteness of a kernel.